EXPERIMENT-3

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USER INTERFACE AND DESIGN

Develop and compare CLI, GUI, and Voice User Interfaces (VUI) for the same task and assess user satisfaction.

1. COMMAND LINE INTERFACE

Python code:

```
import os
import sys
def rename_file(old_name, new_name):
try:
os.rename(old_name, new_name)
print(f"File renamed from {old_name} to {new_name}")
except FileNotFoundError:
print(f"Error: {old_name} not found.")
except Exception as e:
print(f"An error occurred: {e}")
if _name___== "_main_":
if len(sys.argv) != 3:
print("Usage: python rename_file_cli.py <old_filename> <new_filename>")
else:
rename_file(sys.argv[1], sys.argv[2])
```

Output:

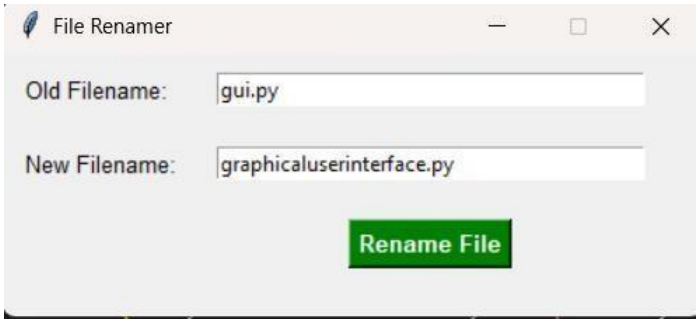
```
PS E:\uid> python -u "e:\uid\cli.py"
Usage: python cli.py <old_filename> <new_filename>
PS E:\uid> python cli.py cli.py commandlineinterface.py
File renamed from 'cli.py' to 'commandlineinterface.py'
PS E:\uid>
```

2. GRAPHICAL USER INTERFACE

```
import tkinter as tk

from tkinter import messagebox

import os

def rename_file():

old_name = old_filename_entry.get().strip()

new_name = new_filename_entry.get().strip()

if not old_name or not new_name:

messagebox.showwarning("Warning", "Please enter both filenames!")

return

if not os.path.exists(old_name):

messagebox.showerror("Error", f"File '{old_name}' not found.")

return

if os.path.exists(new_name):

if not messagebox.askyesno("Warning", f"'{new_name}' already exists. Overwrite?"):

return

try:
```

```
os.rename(old_name, new_name)

messagebox.showinfo("Success", f"File renamed from '{old_name}' to '{new_name}'")

except Exception as e:

messagebox.showerror("Error", f"An error occurred: {e}")

root = tk.Tk()

root.title("File  Renamer")

root.geometry("400x150")

root.resizable(False, False)

tk.Label(root, text="Old Filename:", font=("Arial", 10)).grid(row=0, column=0, padx=10,
pady=5, sticky="w")

tk.Label(root, text=" New Filename:", font=("Arial", 10)).grid(row=1, column=0, padx=10,
pady=5, sticky="w")

old_filename_entry = tk.Entry(root, width=40)

old_filename_entry.grid(row=0, column=1, padx=10, pady=5)

new_filename_entry = tk.Entry(root, width=40)

new_filename_entry.grid(row=1, column=1, padx=10, pady=5)

rename_button = tk.Button(root, text=" Rename File", bg="green", fg="white",
font=("Arial", 10, "bold"), command=rename_file)

rename_button.grid(row=2, column=0, columnspan=2, pady=10)

root.mainloop()
```

Output:



3. VOICE USER INTERFACE

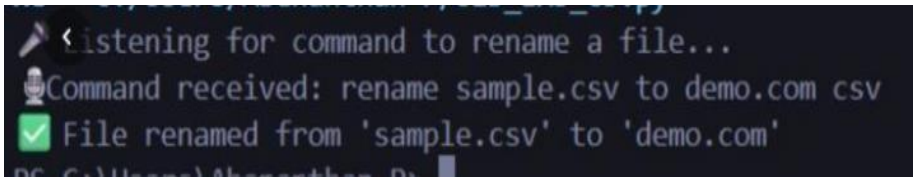
Python Code:

```
import speech_recognition as sr

import os

def rename_file_from_voice_command(command):

try:

words = command.lower().split(" ")

if "rename" in words and "to" in words:

rename_index = words.index("rename")

to_index = words.index("to")

old_name = words[rename_index + 1]

new_name = words[to_index + 1]

if not os.path.exists(old_name):

print(f"Error: File '{old_name}' not found.")

return

os.rename(old_name, new_name)

print(f" File renamed from '{old_name}' to '{new_name}'")
```

```
except Exception as e:

print(f"Error: {e}")

def listen_for_command():

recognizer = sr.Recognizer()

mic = sr.Microphone()

print(" Listening for command to rename a file...")

with mic as source:

recognizer.adjust_for_ambient_noise(source)

audio = recognizer.listen(source)

try:

command = recognizer.recognize_google(audio)

print(f" Command received: {command}")

rename_file_from_voice_command(command)

except sr.UnknownValueError:

print("Sorry, I couldn't understand the command.")

except sr.RequestError as e:

print(f" Could not request results from Google Speech Recognition service; {e}")

if _name___== "_main_":

listen_for_command()
```

Output:

```
PS E:\uid> python -u "e:\uid\vui.py"
Listening for command to rename a file...
Sorry, I couldn't understand the command.
```



4. USER SATISFACTION COMPARISON

Python Code:

```
def survey():
    print("Rate your satisfaction with the following interfaces (1-5):")
    try:
        cli_satisfaction=int(input("CLI(1-5):"))
        gui_satisfaction=int(input("GUI(1-5):"))
        vui_satisfaction=int(input("VUI(1-5):"))
        if not(1<=cli_satisfaction<=5 and 1<=gui_satisfaction<=5 and 1<=vui_satisfaction<=5):
            print("Please enter a number between 1 and 5 for each interface.")
            return
        print("\nYour satisfaction ratings:")
        print(f"CLI: {cli_satisfaction}")
        print(f"GUI: {gui_satisfaction}")
        print(f"VUI: {vui_satisfaction}")
        avg_satisfaction=(cli_satisfaction+gui_satisfaction+vui_satisfaction)/3
        print(f"Average Satisfaction: {avg_satisfaction:.2f}")
    except ValueError:
        print("Invalid input. Please enter a number between 1 and 5 for each interface.")
if __name__=="__main__":
    survey()
```

Output:

```
PS E:\uid> python -u "e:\uid\tempCodeRunnerFile.py"
Rate your satisfaction with the following interfaces (1-5):
CLI(1-5):3
GUI(1-5):4
VUI(1-5):2

Your satisfaction ratings:
CLI: 3
GUI: 4
VUI: 2
Average Satisfaction: 3.00
```

Result:

Thus , the Command Line Interface , Graphical User Interface , Voice Command Interface and User Satisfactory scores to rename a file is created and implemented successfully.